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Dynamic DNA Origami Device for Measuring Compressive Depletion Forces

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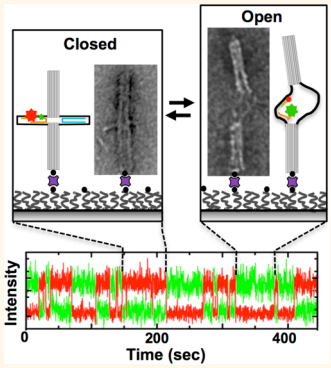
Supporting Information

ABSTRACT: **The ability to self-assemble nanodevices with programmed structural dynamics that can sense and respond to the local environment could enable transformative applications in fields including molecular robotics, nanomanufacturing, and nanomedicine. The responsive function of biomolecules is often driven by alterations in conformational distributions mediated by highly sensitive interactions with the local environment. Here, we mimic this approach by engineering inherent nanoscale structural dynamics (nanodynamics) into a DNA device that exhibits a distribution of conformations including two stable states separated by a transition state where the energy barrier height is on the scale of the thermal energy, k_BT = 4.1 pN·nm, enabling spontaneous transitions between states. We further establish design principles to regulate the equilibrium and kinetic behavior by substituting a few DNA strand components. We use single-molecule Förster resonance energy transfer measurements to show these nanodynamic properties are sensitive to sub-piconewton depletion forces in the presence of molecular crowding agents, and the device can measure depletion forces with a resolution of ∼100 fN. We anticipate that this approach of engineering nanodynamic DNA devices will enable molecular-scale systems that sense and respond to their local environment with extremely high sensitivity.**



KEYWORDS: *DNA nanotechnology, DNA origami, conformational dynamics, molecular crowding, depletion forces, single-molecule biophysics*

The responsive function of many biomolecules is enabled by exhibiting a distribution of conformational states that can be altered by subtle cues in the local environment such as force or binding of a cofactor. For example, the function of heatshock protein 90[1] and the enzyme phosphoglycerate kinase[2] can be activated or enhanced by a redistribution of conformational states driven by molecular crowding. Engineering nanodevices with these well-controlled structural dynamics that respond to their local environment with high sensitivity could enable a powerful approach to probe physical interactions such as crowding forces at a molecular scale. Here we report a DNA nanodevice that exhibits a programmable distribution of states including two stable states separated by an energy barrier. In particular, we exploit the precise design enabled by DNA origami[3] to achieve a device with large conformational changes that are easily detectable with tunable transition rates and state distributions. We further demonstrate the distribution of states measured by single-molecule Förster resonance energy transfer (smFRET) serves as a reporter of compressive depletion forces with resolution of ∼100 fN (∼10 ndyn).

Structural DNA nanotechnology[4] has enabled the design of biomolecular structure with nanometer[3,5,6] or recently even sub-nanometer resolution.[7] Advances in the development of dynamic DNA nanostructures[8−14] have led to devices that can reconfigure between discrete states actuated by external inputs such as the addition of DNA strands or buffer exchange to trigger structural changes on the time scale of ∼10 s or longer.[8,10,13−16] Recent studies have also illustrated nanodevices that exhibit conformational dynamics on the second and even subsecond time scale.[17] The goal of this study was to design dynamic nanostructures exhibiting a tunable ensemble of states that is responsive to physical interaction with the local environment to enable a stand-alone device that reports on local forces. To achieve this goal, we developed a nanodevice that incorporates weak energy barriers (scale of k_BT) to regulate conformational dynamics between multiple stable states on the second time scale. This time scale allows for relatively fast response but straightforward measurement with smFRET. Specifically, the device fluctuates between compact and open states. Our design strategy enables programming configurational entropy to tune the relative free

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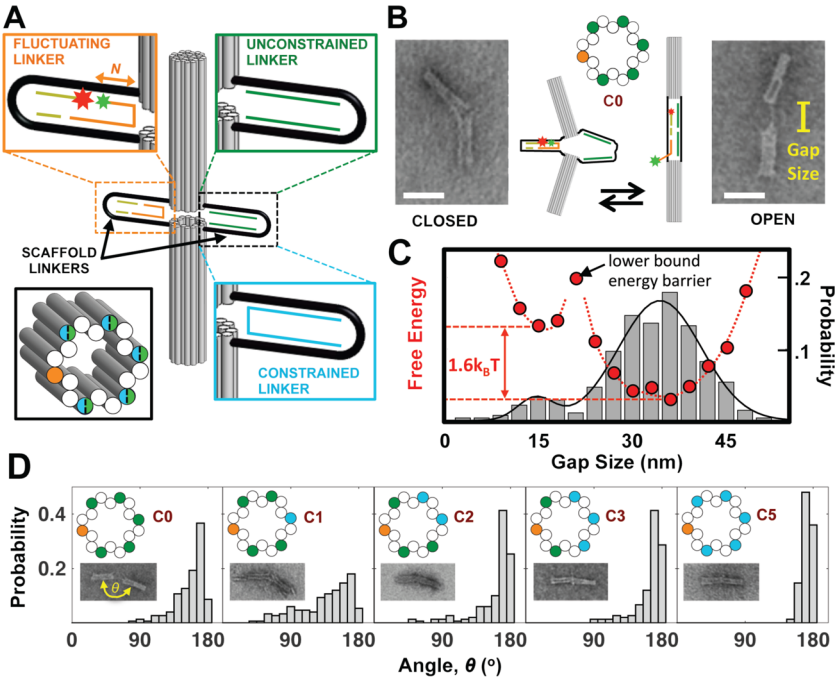

Figure 1. Design of the NanoDyn device. (a) The NanoDyn contains two large DNA barrels connected by a fluctuating scaffold linker (orange) and five additional scaffold linkers (green or cyan) that can be different combinations of unconstrained (green) and constrained (cyan) linkers to control the conformational space. Only two scaffold linkers are shown for clarity; however all six are shown in the cross-section. (b) Schematic and TEM image for a closed and open NanoDyn with no constraints (C0); scale bars = 50 nm. (c) Free energy landscape and probability histogram for a C0 NanoDyn determined from the probability distribution of the gap size between the two cylinders of the C0 NanoDyn device ($N = 276$). The probability histogram was fit to the sum of two Gaussian distributions with average gap sizes of 15 ± 4 nm and 34 ± 9 nm (black line). The probability at the transition state is an upper bound, and hence the barrier height is a lower bound. (d) Angular probability of the C0, C1, C2, C3, and C5 NanoDyn devices. Insets show the unconstrained (green) and constrained linkers (cyan) on the cross-section and TEM images with an angle similar to the average for each constraint condition.

energy difference between these states. We independently quantified this free energy difference with transmission electron microscopy (TEM), bulk fluorescence, and smFRET measurements. Ensemble fluorescence and smFRET measurements also revealed thermally driven structural dynamics with tunable equilibrium and dynamic properties that are sensitive to sub-piconewton molecular crowding forces generated by poly(etheylene glycol) (PEG) crowding molecules.

RESULTS AND DISCUSSION

Nanodynamic Device Design and Static Characterization. Our scaffolded DNA origami device (Figure 1a, Supplemental Table S1, Supplemental Figure S1), which we refer to as a NanoDyn due to its nanodynamic behavior and its sub-piconewton force resolution (10 fN = 1 ndyn), has three key functional characteristics: (i) thermally driven conformational dynamics on the second time scale, (ii) tunable transition rates, and (iii) sensitivity to forces from the local environment. The device design comprises two barrel components and adopts either a "closed state", which is an ensemble of conformations where base pairing interactions hold the barrels close to each other, or an "open state", which is an ensemble of conformations where the six scaffold linkers allow significant relative motion of the barrels (Figure 1b, only two linkers are shown for clarity). Conformational changes between the closed and open ensembles are detected *via* TEM and smFRET. The NanoDyn is closed *via* base pairing of a single strand to the fluctuating linker that converts it into a loop (Figure 1a, top left). We chose one of the loop binding sites to be a 10 base pair closing interaction because, similar to DNA-PAINT methods,[18,19] the binding is transient, giving dissociation dynamics on the second time scale that could be accurately detected with smFRET. In the baseline design, the other five scaffold linkers are in the "unconstrained" state (Figure 1a, top right), where they act like a flexible hinge that allows for significant relative motion between the two barrels. TEM images revealed homogeneous well-folded devices exhibiting a range of conformations (Supplemental Figure S2). Figure 1c depicts the equilibrium conformational distribution of the unconstrained NanoDyn, quantified by TEM in terms of the gap size between the two components. Ninety-two percent of devices appear in the open state (determined from Gaussian fits), where the barrels are clearly separated. The structure hinges open (Figure 1b, left) yielding a gap size distribution with a standard deviation of 4 nm and mean of 15 ± 2 nm (±95% confidence interval from the Gaussian fit). In the open state the barrels are generally separated by a larger gap exhibiting a gap size with a standard deviation of 9 nm and a mean of 34 ± 1 nm (±95% confidence interval from the Gaussian fit). These measurements agreed well with the gap sizes for NanoDyn devices that were all open or all closed, respectively (Supplemental Figure S3). To analyze the free energy difference between the open and closed ensembles, the gap size probability, P (Figure 1c), was converted to a free energy landscape using the Boltzmann distribution where the relative free energy is $-k_BT \ln(P)$, revealing a free energy difference between the closed and open states $G_{closed} - G_{open} = \Delta G_{closed\text{-}open} = 1.6\ k_BT$ and a free energy barrier on the scale of k_BT. While we can resolve the presence of an energy barrier, there is likely a lower probability transition state that we cannot resolve due to the limited resolution of our TEM measurements. Hence the height of the energy barrier quantified

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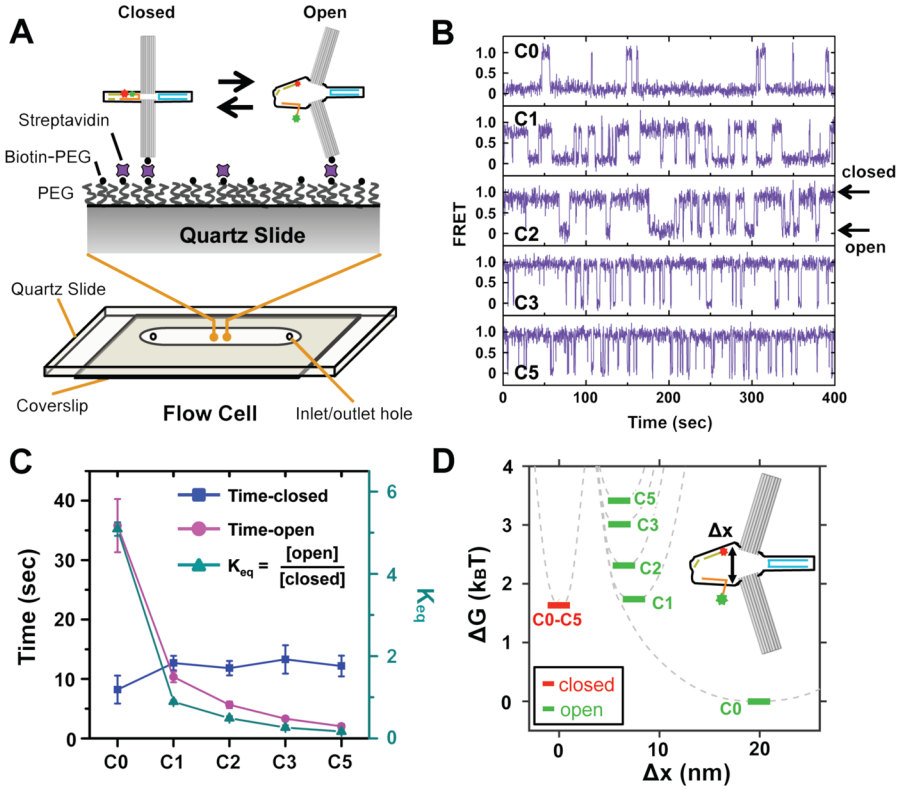

Figure 2. smFRET measurements of the NanoDyn devices. (a) Schematic of the microscope flow cell with a NanoDyn device tethered to the functionalized slide surface (see Methods). (b) Example single-molecule FRET time series of C0−C5 NanoDyn devices showing fluctuations between high-FRET (closed) and low-FRET (open) states. (c) Average dwell time of the C0−C5 NanoDyn devices in the open and closed state (left axis) and the equilibrium constant (right axis). Error bars represent the standard deviation of the average values from three randomly divided subgroups of equal sizes. (d) Schematic energy landscapes for C0−C5 NanoDyn devices showing the relative energy difference (ΔG) as a function of the reaction coordinate (Δx), depicted in the inset. We assumed the energy well of the closed state (red) is the same for each structure. The closed state of the C0 NanoDyn was chosen as the 0 k_BT reference point, and relative ΔG values come from the data presented in Table 1.

from our probability is a lower limit to the energy required to transition between states. Since we cannot accurately resolve the transition state, we represented the energy landscape as two separate energy wells (Figure 1c).

Controlling Conformational Space of Open NanoDyn Devices. We developed a strategy to tune the free energy difference between the open and closed states by reducing the entropy of the open state, which effectively decreases the energy difference between open and closed states. The five unconstrained linkers can be modified to reduce the configurational space of the open ensemble by incorporating a single staple that forms a stable scaffold loop, pinning the barrels close together (Figure 1a, bottom right). We define a parameter, *C*, which indicates the number of constrained scaffold linkers, corresponding to the number of points around the perimeter where the barrels are stably pinned together. We fabricated NanoDyn devices (Figure 1d) with constraints ranging from C0 (no constrained linkers) to C5 (five constrained linkers). Gel electrophoresis and TEM imaging revealed well-folded structures with notable differences in conformation depending on the constraint number (Supplemental Figure S4). The unconstrained C0 NanoDyn typically had a large gap size (Figure 1c and Supplemental Figure S3), indicating a large configurational space, with a slight tendency to align (Figure 1d). The addition of one constraint staple (C1) yielded significantly smaller gaps (Supplemental Figure S3) and caused the device to kink, resulting in a broad angular distribution (Figure 1d). For the C1 NanoDyn, the barrels are pinned together at one point on the perimeter and are primarily constrained to relative rotational motion. Since they are pinned together at the perimeter, the gap size at the center is nonzero. Pinning the barrels together at more points about the perimeter decreases the range of rotation angles and likely suppresses rotational degrees of freedom. C2 and C3 devices appeared primarily closed on TEM images (Supplemental Figure S4) with the two barrels strongly aligned. All C5 NanoDyn devices appeared to be closed and exhibited nearly straight barrel alignment (Figure 1d and Supplemental Figure S3). The gap size was too small to reliably measure by TEM analysis for C2, C3, and C5 devices. Combined, these results suggest that a single constraint, C1, suppresses translational motion, and the addition of C2 through C5 constraints suppresses rotational motions, with large changes in configurational space occurring from C0 to C1 (translational) and C1 to C2 (rotational) and only minor changes from C2 to C3 and C5.

Single-Molecule Characterization of NanoDyn Structural Dynamics. We used smFRET measurements[20] to further characterize the equilibrium and dynamic properties of the NanoDyn (Figure 2a). Representative smFRET traces (Figure 2b) (C0 through C5) display high and low FRET states corresponding to closed and open devices, respectively. The C0 NanoDyn exhibits an equilibrium constant of K_{eq} = 5.1 ± 0.2, corresponding to 84% open (details of K_{eq} and energy calculations in the Supporting Information). This agrees very well with fraction open of 92% from our TEM measurements (Figure 1c), suggesting our TEM measurements are an accurate depiction of NanoDyn equilibrium behavior in solution and that

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Table 1. Thermodynamic and Kinetic Parameters of NanoDyn with Different Constraints (C0−C5)

sample	t_{closed} (s)	t_{open} (s)	fraction open	K_{eq}	$\Delta G_{closed\text{-}open}$ (k_BT)	$\Delta\Delta G_{C0\text{-}CN} = \Delta G_{C0} - \Delta G_{CN}$ (k_BT)	$\Omega_{rel,C0}/\Omega_{rel,CN}$
C0	8 ± 2	36 ± 4	0.84 ± 0.03	5.1 ± 0.2	1.63 ± 0.03	0.00 ± 0.05	1
C1	13 ± 1	10.3 ± 0.9	0.47 ± 0.02	0.89 ± 0.03	−0.12 ± 0.04	1.74 ± 0.05	5.7 ± 0.3
C2	12 ± 1	5.7 ± 0.6	0.33 ± 0.04	0.49 ± 0.05	−0.7 ± 0.1	2.3 ± 0.1	10 ± 1
C3	13 ± 2	3.3 ± 0.3	0.21 ± 0.05	0.26 ± 0.06	−1.3 ± 0.2	3.0 ± 0.2	19 ± 4
C5	12 ± 2	2.0 ± 0.4	0.14 ± 0.04	0.17 ± 0.04	−1.8 ± 0.3	3.4 ± 0.3	31 ± 8

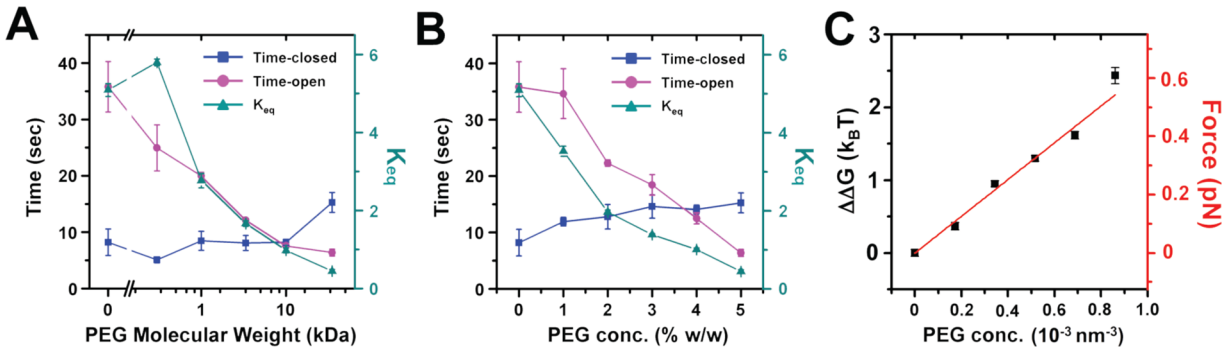

Figure 3. Dynamic behavior of C0 NanoDyn in the presence of PEG. Average dwell time of NanoDyn devices in the open and closed state (left axis) and the equilibrium constant (right axis) for (a) C0 NanoDyn devices as a function of PEG molecular weight at 5% w/w concentration. (b) C0 NanoDyn devices as a function PEG_{35} concentration. (c) PEG_{35} induced changes in the free energy difference ($\Delta\Delta G_{PEG}$) between the open and closed C0 NanoDyn devices (left axis) is a linear function. The depletion force (right axis) determined from $\Delta\Delta G_{PEG}/\Delta L$ reveals an approximately femtonewton scale force measurement resolution.

the opening/closing of the fluctuating linker, which is read out by smFRET, is well-correlated to the overall opening/closing of the gap as measured by TEM. Additional constraints (C1, C2, C3, C5) sequentially reduce the K_{eq}, where the C5 NanoDyn has a K_{eq} = 0.17 ± 0.04, corresponding to 14% open (Figure 2c and Table 1). Equilibrium constants determined from ensemble measurements of FRET efficiency agree well with the smFRET measurements (Supplemental Figure S4), indicating that surface tethering does not significantly influence the smFRET measurements.

Figure 2d illustrates energy landscapes reconstructed from the smFRET results (Table 1), assuming the closed state free energy is similar for all NanoDyn devices. Since we are concerned with energy changes, we set the free energy for the open C0 NanoDyn at zero. Since we determined only a lower bound to the energy barrier height, we did not include the transition state in this energy landscape description. The Δx value for C0 and C1 NanoDyn devices were estimated from the TEM measurements, while the higher constraints were assumed to gradually decrease from C1. Although the true energy landscapes are likely more complicated, the dashed gray lines represent a simple energy landscape that captures the ΔG values as a guide to conceptualize the structural dynamics. Since the C0 through C5 NanoDyn devices all contain the same 10 bp closing interaction, the difference in enthalpy between the open and closed states is similar for the C0 through C5 devices. Using this assumption, we calculated the relative conformational space between the open and closed states (*i.e.*, the ratio of the number of accessible states, $\Omega_{rel} = \Omega_{open}/\Omega_{closed}$) based on the experimentally determined free energy difference. Our results (Table 1) imply that the C1 constraint reduces the conformational space by 6-fold. The additional C2 and C3 constraints reduce the conformational space by an additional 2-fold each, while the combined addition of the fourth and fifth constraints reduces the NanoDyn conformational space by a further 1.5-fold. Since the gap size varies significantly for only the C0 NanoDyn, the entropic difference between the C0 and C1 NanoDyn is likely due to limiting translational freedom, while the additional constraints (C2−C5) gradually suppress angular conformational space (Figure 1d).

Kinetic Properties of NanoDyn Devices with Varying Constraints. We determined kinetic properties of the C0−C5 NanoDyn devices from the dwell time distributions of the open and closed states, which revealed two characteristic transition time scales for both opening and closing transitions (Supplemental Figure S5). This could be due to a number of reasons including secondary interactions that do not impact the FRET state, but may occur with low probability and influence some of the dwell times. A second possibility could be inhomogeneity of NanoDyn devices; however, structures appeared to be highly uniform in TEM images (Supplemental Figures S2−S4). Furthermore, plotting average dwell times for individual NanoDyn devices did not reveal any clear subpopulations (Supplemental Figure S6). Another possibility could be that different types of motion impact the transitions such that multiple reaction coordinates are necessary to describe the state of the device. Interestingly, we observed that the characteristic dwell times in the closed state are unaffected by the constraint conditions changing from C0 through C5 (Supplemental Figure S7). In contrast, the longer characteristic open state dwell time exhibited a large change from C0 to C1 and then remained relatively constant from C1 to C5. This may suggest the longer open state dwell times are related to large translational motions of the barrels. The shorter characteristic open state dwell time systematically decreased with increasing constraint (Supplemental Figure S7). Fully understanding the source of these multiple characteristic time scales will likely require probing additional reaction coordinates in a multidimensional free energy landscape. However, since our primary goal is to demonstrate an ability to tune kinetic behavior, we used the average dwell time (t_{open} and t_{closed}) as a model-free quantification of the dynamics (Table 1). The constraints iteratively reduced t_{open} from 36 ± 4 s

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Table 2. Thermodynamic and Kinetic Parameters of C0 NanoDyn with Increasing Concentration of 35 kDa PEG

$[PEG_{35}]$ (%)	$[PEG_{35}]$ (mM)	t_{closed} (s)	t_{open} (s)	fraction open	K_{eq}	$\Delta G_{closed\text{-}open}$ (k_BT)	$\Delta\Delta G_{PEG} = \Delta G_{no\ PEG} - \Delta G_{PEG}$ (k_BT)	ΔV_{ex} (nm^3)	depletion force (pN)
0	0	8 ± 2	36 ± 4	0.84 ± 0.03	5.1 ± 0.2	1.63 ± 0.03	0.00 ± 0.05		
1	0.28	11.9 ± 0.7	35 ± 4	0.78 ± 0.03	3.5 ± 0.1	1.26 ± 0.04	0.37 ± 0.05	2100 ± 300	0.09 ± 0.01
2	0.57	13 ± 2	22.3 ± 0.6	0.66 ± 0.02	1.97 ± 0.05	0.68 ± 0.02	0.95 ± 0.04	2800 ± 100	0.24 ± 0.02
3	0.86	15 ± 2	18 ± 2	0.58 ± 0.01	1.39 ± 0.03	0.33 ± 0.02	1.30 ± 0.04	2510 ± 70	0.32 ± 0.02
4	1.14	14.1 ± 0.8	12 ± 1	0.50 ± 0.02	1.01 ± 0.04	0.01 ± 0.04	1.62 ± 0.05	2350 ± 80	0.40 ± 0.03
5	1.43	15 ± 2	6.4 ± 0.6	0.31 ± 0.03	0.45 ± 0.05	−0.8 ± 0.1	2.4 ± 0.1	2800 ± 100	0.61 ± 0.05

(C0) to 2.0 ± 0.4 s (C5) (Figure 2c), while only the first constraint increased t_{closed}, which changed from 8 ± 3 s (C0) to 13 ± 1 s (C1). This indicates that this reduction in conformational space increases the open state free energy, which reduces the free energy barrier to closing and shifts the equilibrium toward the closed state (Figure 2d). Our design allows for tuning of the closing rate and equilibrium constant by an order of magnitude through the substitution of a few DNA staples.

Nanodynamics Are Sensitive to Molecular Crowding Forces. We hypothesized that the equilibrium and transition rates of our NanoDyn device with energy barriers comparable to k_BT would be highly sensitive to applied forces. We focused on compressive depletion forces due to molecular crowding, which influence numerous biomolecular processes including actin bundle formation, enzyme activity, and molecular motor processivity[21−24] and have been shown to mediate protein activity by shifting conformational distributions,[1,2] but are challenging to quantitatively measure at the molecular scale. We reasoned that in a crowded environment the additional excluded volume of the open NanoDyn reduces the entropy of the crowding molecules, thereby biasing the device toward the closed state as reported for a protein-based device.[25] We initially focused on smFRET measurements of the C0 NanoDyn, since they exhibited the longest open state dwell times, and used PEG as the crowding molecule because of its wide use[24,25] and biocompatibility.

We first investigated the C0 NanoDyn equilibrium and kinetic properties in 5% w/w solution of PEG with increasing molecular weight (MW) (Figure 3a, Supplemental Figures S8 and S9, Supplemental Table S2). All PEG molecular weights except 0.3 kDa ($PEG_{0.3}$) decreased the C0 NanoDyn K_{eq}. This indicates the C0 NanoDyn excluded volume is larger in the open state than the closed state for all PEG molecular weights except 0.3 kDa. This suggests that $PEG_{0.3}$ is not excluded from the gap in either state. Using a previously derived scaling law for the radius of gyration, $R_g = 0.215 \times M_w^{0.583}$,[26] we estimated effective sizes (twice the radius of gyration) of 1.2, 2.4, 4.6, 9.2, and 19.2 nm for the $PEG_{0.3}$, PEG_1, PEG_3, PEG_{10}, and PEG_{35}, respectively. The spacing between linkers at the point where they connect to the barrels is 4 nm, which is consistent with the $PEG_{0.3}$ being able to enter the gap and excluding the PEG_3, PEG_{10}, and PEG_{35}. Although the PEG_1 may be smaller than this spacing on average, fluctuations of the linkers and extended fluctuations of the PEG_1 polymers (contour length of PEG_1 is ∼8 nm) may still exclude them from the gap. In contrast to C0, we find the C1 NanoDyn equilibrium is reduced by $PEG_{0.3}$ (Supplemental Figures S10−12, Supplemental Table S3), suggesting that, by constraining the size of the gap between the two barrels, the difference in excluded volume between the open and closed NanoDyn becomes sensitive to smaller molecules and illustrates how we can tune the measurement sensitivity through structure design parameters.

We next investigated the influence of increasing depletion forces on the C0 and C1 NanoDyn dynamics by varying the concentration of PEG_{35} and PEG_1, respectively (Figure 3b, Table 2, Supplemental Figures S13−S19, Supplemental Table S4). We focused further analysis on the C0 device since the conformational change is primarily translational, making it easier to determine changes in excluded volume and depletion forces. The devices exhibited two characteristic time scales for transitions in the presence of PEG, with the longer characteristic dwell time in the open state appearing the most sensitive to the presence of PEG (Supplemental Figure S16), which is consistent with the longer dwell times in the open state being related to relatively large translational motions of the barrels. For consistency we again used the average dwell times as a model-free measure of the dynamics (Figure 3b). The C0 NanoDyn K_{eq} decreases with increasing PEG_{35} concentration, again largely due to faster closing (*i.e.*, decreased t_{open}). In this case, there is a systematic increase in t_{closed}, which is consistent with the depletion force-inhibiting opening of the device.

The NanoDyn Free Energy Difference Quantitatively Detects Depletion Forces. We find that the change in $\Delta G_{closed\text{-}open}$ induced by the addition of PEG, $\Delta\Delta G_{PEG}$, depends linearly on the concentration of PEG_{35} (Figure 3c). This is consistent with an entropically induced $\Delta\Delta G_{PEG} = k_BT \times C_{PEG} \times \Delta V_{ex}$,[27] where ΔV_{ex} is the difference in excluded volume between the open and closed states. We determined that ΔV_{ex} is 2520 ± 80 nm^3 for the C0 NanoDyn from the slope of $\Delta\Delta G_{PEG}$ *vs* C_{PEG} (Figure 3c, See methods). To then determine the change in length, ΔL, we estimate the change in excluded volume as $\Delta V_{ex} = A_{barrel} \times \Delta L$, where A_{barrel} is the cross-sectional area. The diameter of dsDNA helices has been observed to effectively swell to ∼2.0−2.5 nm in DNA origami nanostructures,[28] which corresponds to a cross-sectional area of A_{barrel} = 140 ± 30 nm^2. Dividing ΔV_{ex} by this A_{barrel} gives a change in the gap size between the open and closed states as ΔL = 18 ± 4 nm. This agrees well with the ΔL of 19 ± 3 nm that was determined from TEM gap size analysis (Figure 1c).

The depletion force acting to close the C0 NanoDyn at a given C_{PEG} can be calculated from $\Delta\Delta G_{PEG}/\Delta L$ (Figure 3c and Table 2). Therefore, we can consider the NanoDyn as a force sensor, with the measured $\Delta\Delta G_{PEG}$ serving as a readout of the applied force. Our measurement error in $\Delta\Delta G_{PEG}$ with no PEG is ±0.05 k_BT (Table 2). Considering that reliable detection requires a signal that is 3 times the measurement error, the NanoDyn can reliably detect a minimum $\Delta\Delta G_{PEG}$ of 0.15 k_BT. Assuming a similar uncertainty of 0.05 k_BT at this minimum detection limit and taking the length change as ΔL = 18 ± 4 nm, the minimum depletion force that we could detect is F_{depl} = 40 ± 10 fN. The NanoDyn depletion force sensing behavior can be defined by $\Delta\Delta G_{PEG} = F_{depl} \times \Delta L$. Our ability to design a large ΔL of ∼18 nm, which is larger than the full size of most biomolecules, yields a measurement sensitivity of ∼4 k_BT/pN. Assuming a similar

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measurement uncertainty, the detection limit would scale inversely with ΔL, meaning a typical size biomolecule (~5 nm) could likely respond only to forces that are at least several hundreds of femtonewtons using this dynamic sensing approach.

CONCLUSIONS

Here we demonstrated single-molecule measurement and control of DNA nanostructure equilibrium properties and kinetics at the second time scale. We determined a free energy difference between the open and closed states of the NanoDyn of 1.6 k_BT, which was remarkably consistent across measurements using TEM, smFRET, and bulk fluorescence methods. The good agreement between TEM and fluorescence results indicates the open or closed state of the device is not affected by the surface deposition used in TEM studies. This is consistent with prior work demonstrating conformations of DNA polymers are kinetically trapped on surfaces with strong adsorption such as glow-discharged surfaces.[29] In addition, the length change measured from the gap size distribution agrees well with the length change determined from the depletion force measurements. Overall, the combination of these results suggests key equilibrium parameters can be reliably determined from TEM, smFRET, and bulk FRET methods.

This work establishes a foundation for DNA origami devices that can achieve high-sensitivity measurement functions based on thermally driven mechanical dynamics that respond to the local environment. This class of devices enables programming an ensemble of states including states with tunable configurational entropy, leading to control over equilibrium and kinetic properties. We can tune the device equilibrium and kinetics by more than an order of magnitude through engineering the device entropy by substituting a few DNA oligonucleotides. This provides a distinct approach to design DNA nanodynamics. Instead of switching between states of a DNA nanostructure *via* DNA input strands[10,13−15,30] or base stacking interactions,[8] we program a distribution of states and rely on thermal fluctuations to interconvert between states with kinetics on the second time scale that are sensitive to external forces.

Our binary readout of structural dynamics (*i.e.*, open/closed) that is quantified with equilibrium constants and average dwell times provides exquisite force sensitivity that is largely defined by the ability to design a large change in excluded volume. Our design allowed us to detect forces down to 40 fN. Measuring similar forces by the typical approach of directly measuring deformations would require a sensor stiffness of $k_s \approx 10-20$ fN/nm assuming a typical FRET working range of 5−10 nm. This low stiffness would yield thermal fluctuations of $\langle x^2 \rangle^{1/2} = k_BT/k_s \approx 14-20$ nm, making detection of such small forces by measuring deformation extremely challenging. Further, detecting small changes in smFRET is itself highly challenging. Our approach allows for femtonewton scale forces to be detected by large changes in smFRET. However, detecting these forces *via* NanoDyn dynamic behavior requires averaging over time, implying that the measurement temporal resolution is a key consideration. Faster measurements could be achieved by changes in the NanoDyn device that reduce the open and closed dwell times down to the millisecond range that could be detected with ~50 Hz resolution using EMCCD detection[20] and potentially even sub-millisecond using APDs with kHz resolution.[31] Furthermore, the ability to quantitatively probe molecular scale depletion forces could enable critical insight into physical processes in crowded environments such as inside cells, where depletion forces influence processes such as cytoskeletal bundle formation,[22] binding of DNA to DNA polymerase,[32] or assembly of proteins and protein complexes.[33]

Recently, a protein-based FRET sensor was reported for sensing depletion forces within cells.[25] This study connected two fluorescent proteins, mCerulean and mCitrine, through a flexible hinge and detected depletion forces *via* an increase in the average FRET efficiency. This protein engineering approach has the main advantage that it can be genetically encoded into cells. However, smaller DNA nanodevices have been used in cells,[34,35] and recent work introduced methods for electrotransfection of DNA origami into cells,[36] providing a pathway to cellular implementation. The main advantages of the NanoDyn are that the device design and readout provide a quantitative measure of depletion forces as opposed to just a relative measure. In addition, design changes to control functional characteristics such as sensitivity and detection limit are significantly easier to engineer than a protein-based device. Finally, our use of Cy3−Cy5 fluorophores instead of fluorescence proteins enables single-molecule characterization, allowing for direct measurement of dynamic behavior.

More broadly, this work provides a foundation for designing DNA nanodevices with controllable thermally driven dynamics where digital conformational state readouts can report highly sensitive physical measurements in nanoscale environments such as complex polymer solutions, nanofluidic systems, or biological systems. Exploiting the ability to transfect DNA origami structures into cells,[36] introducing the NanoDyn or similar devices could enable quantification of crowding forces that play a major role in biomolecular processes. In addition, the NanoDyn device could be viewed as a molecular piston that is opened and closed by changing the PEG concentration. The integration of this device into a microfluidic device that rapidly changes PEG concentrations could continuously cycle the DNA nanopiston where the energy source is the fluctuating PEG concentrations. Finally, this approach of detecting an analog signal with a digital readout, which is used throughout electronics, could serve as a mechanism to couple to other logic, for example, *via* DNA-based molecular computing.[11,37]

METHODS/EXPERIMENTAL

NanoDyn Design and Assembly. NanoDyn devices were designed in caDNAno[38] (Supplemental Figure 1). Each design contains ~165 staples, with the exact number being defined by its constraint C and fluctuating linker length. NanoDyn devices in the absence of a fluctuating linker (N = 0 bp) and a high-affinity fluctuating linker (N = 20 bp) were also folded. NanoDyn devices were folded following previously described protocols.[39] Briefly, the scaffold at 20 nM was combined with a 10-fold excess of each individual staple in the self-assembly reaction buffer containing 1 mM EDTA, 5 mM NaCl, 5 mM Tris, and 18 mM $MgCl_2$. Folding reactions were subjected to a thermal annealing ramp with initial heating to 65 °C followed by a slow cooling process to 4 °C (details in the Supporting Information). NanoDyn devices were purified by gel electrophoresis run at 70 V on a 2% agarose gel with 0.5× TBE, 11 mM $MgCl_2$, and 400 ng/mL ethidium bromide (Supplemental Figure S3a). Well-folded structure bands were extracted from the gel using Freeze 'N Squeeze DNA gel extraction spin columns (Bio-Rad Laboratories, Inc., Hercules, CA, USA).

NanoDyn devices used for ensemble and single-molecule FRET measurements were purified following the protocol described in Stahl *et al.*[40] In summary, a 1:1 volume ratio of folded structures with 15% PEG 8000 and 500 nM NaCl are mixed and spun at 16 000 rcf for 25 min. After centrifugation, the pellet was resuspended in the appropriate buffer. This process was repeated twice, and the final resuspension buffer was 0.5× TBE with 11 mM $MgCl_2$ and 0−5% PEG of varying MW (from 0.3 to 35 kDa).

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ACS Nano 2017, 11, 6566−6573

TEM Imaging and Analysis of NanoDyn Devices. NanoDyn TEM samples were prepared as described in Castro *et al.*[39] Briefly, gel-purified structures at ∼1−5 nM were incubated on a Formvar/carbon square mesh grid (Electron Microscopy Sciences, Hatfield, PA, USA) for 4 min, washed with 10 μL of 2% uranyl formate, then incubated with 20 μL of 2% uranyl formate as a staining agent for 40 s. The stain was then removed by dabbing with filter paper and allowed to dry for at least 30 min prior to imaging. Samples were imaged on an FEI Tecnai G2 Spirit TEM at an acceleration voltage of 80 kV and 39 000−85 000× magnification. Sample TEM images are shown in Supplemental Figure S3b.

ImageJ was used to measure the angle made by the two barrels of the NanoDyn (Figure 1d) as well as the gap length between them (Supplemental Figure S4). The measurement uncertainty for the angle analysis was determined to be 1.9° (Supplemental Figure S4b). For the gap analysis, accuracy was limited to ±3.5 nm (due to a pixel size of 0.7 nm/pixel and a point measurement accuracy of 5 pixels), while measurement error was calculated to be 2.8 nm (SD) by making 50 measurements on the same structure. Because of resolution limits, gap size was detected only for the C0 and C1 NanoDyn devices.

Ensemble Fluorescence Measurements. NanoDyn devices were assembled and purified as described above. Fluorescence emission spectra of each NanoDyn at 10 nM concentration were acquired with a Fluoromax 4 fluorometer (Horiba, Edison, NJ, USA) in a buffer containing 1 mM EDTA, 44.5 mM Tris, 44.5 mM boric acid, and 11 mM $MgCl_2$. Emission spectra were acquired from 530 to 750 nm, while the Cy3 donor fluorophore was excited at 510 nm. Emission spectra were separately acquired from 630 to 750 nm, while the Cy5 acceptor was directly excited at 610 nm. Fluorescence spectra were analyzed in Matlab using the $(\text{ratio})_A$ method[41] to determine the FRET efficiency E_{ff}. The fraction of open devices from the ensemble measurements was calculated as $1 - E_{ff}$. The uncertainty was determined from the standard deviation of the measurement done in duplicate with NanoDyn devices assembled and purified independently.

Single-Molecule FRET Experiments. Single-molecule FRET experiments were carried out with a prism-based TIRF microscope built on an inverted microscope (Olympus, IX71) as previously described.[42,43] Cy3 and Cy5 fluorophores were excited using 532 and 638 nm diode lasers (Crystal Lasers), respectively. A Pellin-Broca prism (Melles Griot) guided the excitation laser light to the quartz microscope slide surface at an incident angle larger than the critical angle for total internal reflection. The fluorescence emission from surface-tethered NanoDyn devices was collected using a 1.2 NA water immersion objective (Olympus, UPlanSApo 60×/1.20w). Separate images of the Cy3 and Cy5 emission were collected using a DualView system (Optical Insights) with a dichroic beam splitter (Chroma Tech, T635lpxr), band-pass filters (Chroma Tech, D585/30 and D680/35), and an EMCCD camera (Princeton Instruments, PhotonMax 512) with Winview software (Princeton Instruments) on a PC.

Flow cells for smFRET experiments were made using quartz microscope slides (G. Finkenbeiner, Waltham, MA, USA) and glass coverslips functionalized with PEG. NanoDyn devices were immobilized on the surface *via* biotin−streptavidin binding and suspended in a final imaging buffer prior to smFRET experiments. Specific details of PEG functionalization and smFRET assay preparation are provided in the Supplementary Methods. For experiments with PEG, the desired molecular weight and concentration were added to the final imaging buffer before injecting it into the flow cell. NanoDyn devices with a Cy5 emission signal under direct excitation by the 638 nm laser, and either a Cy3 signal (low FRET) or Cy5 signal (high FRET) when excited by the 532 nm laser, were considered to have both Cy3 and Cy5 fluorophore labels. Only these molecules with both fluorophores were further analyzed. For each single-molecule time series, 2000 frames of images were acquired at a 5 Hz frame rate for 400 s. Details of the smFRET data analysis are provided in the Supplementary Methods.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsnano.6b07097.

Additional information (PDF)

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Notes

The authors declare no competing financial interest.

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